Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-6)

PHARMACEUTICAL BIOTECHNOLOGY-THEORY

Subject Code : BP-605T

M.Code : 77990

Date of Examination : 02-12-2025

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION - A

1. **Answer Briefly :**

   a) Define enzyme biotechnology.

   b) What is the function of biosensors in pharmaceutical industries?

   c) Name any two microbes used in enzyme production.

   d) What is the role of DNA ligase in rDNA technology?

   e) Mention any two therapeutic proteins produced using genetic engineering.

   f) Define vaccine with an example.

   g) What are plasma substitutes? Give one example.

   h) Expand ELISA and state one application.

   i) What are transposons?

   j) List two fermentation products of pharmaceutical importance.

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SECTION - B

2. Describe in detail the production of industrially important enzymes: amylase, lipase, and protease.

3. Discuss the application of DNA technology in the production of vaccines and insulin.

4. Write detailed notes on ELISA and Western blotting techniques.

SECTION - C

5. Write a note on biosensors and their role in drug discovery.

6. Explain the functions of MHC molecules.

7. Discuss immune stimulation and immune suppression.

8. Write short notes on antitoxins and serum-immune blood derivatives.

9. Explain conjugation in bacteria with a neat diagram.

10. Discuss microbial genetics: plasmids and their applications.

11. Describe large-scale fermenter requirements.

12. Explain the industrial production of Vitamin B12.

13. Write a note on processing and storage of dried human plasma.

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NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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